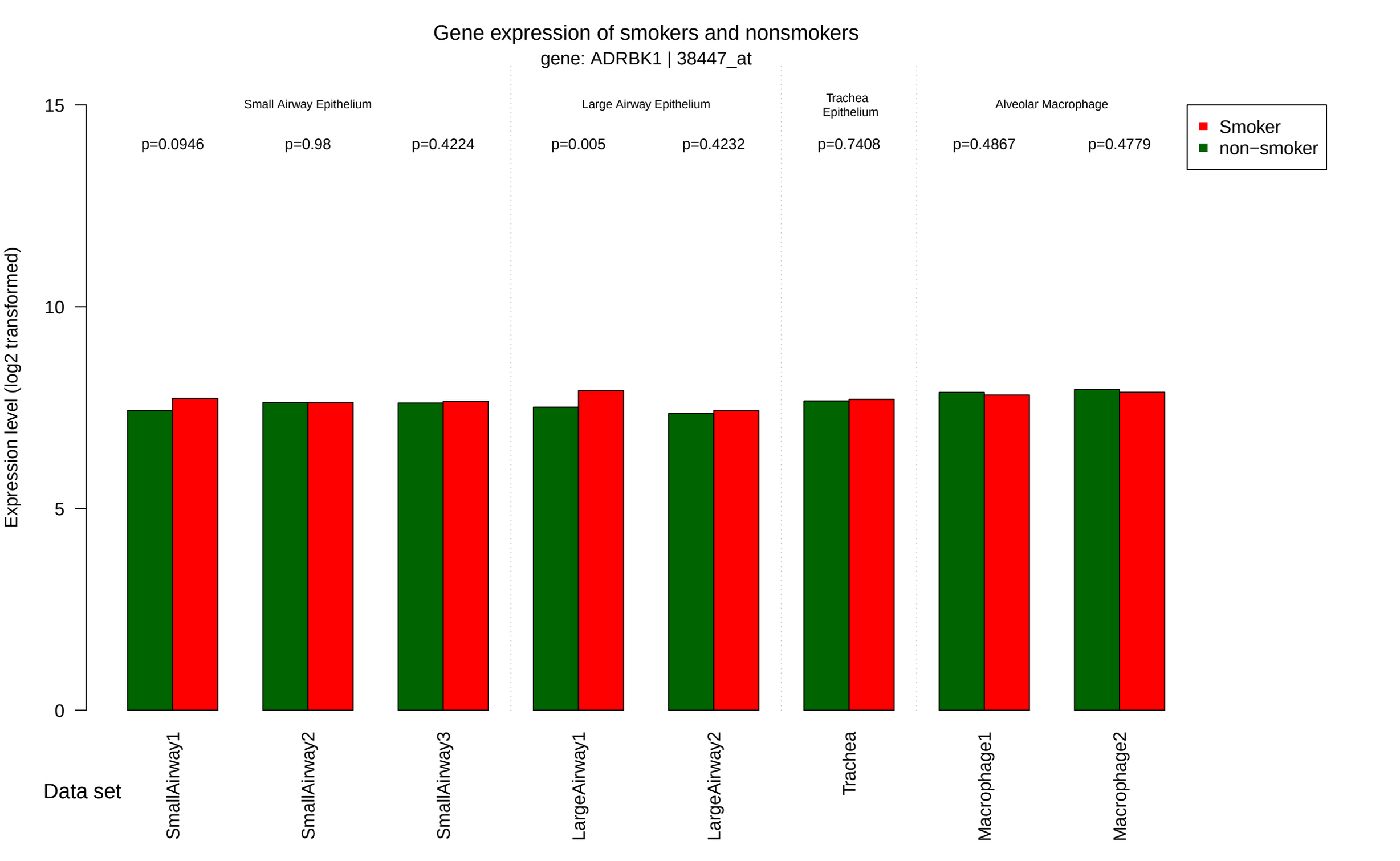
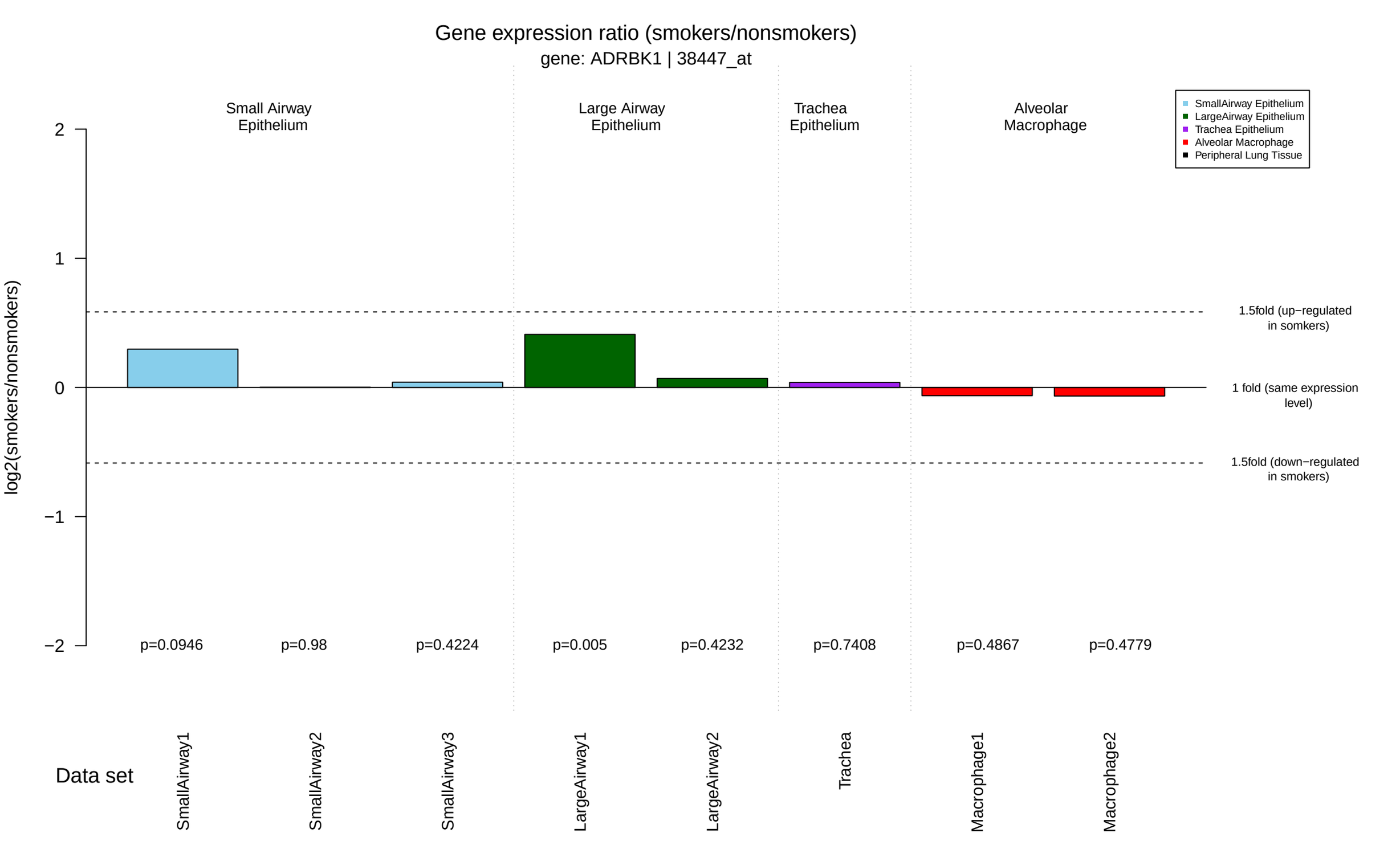
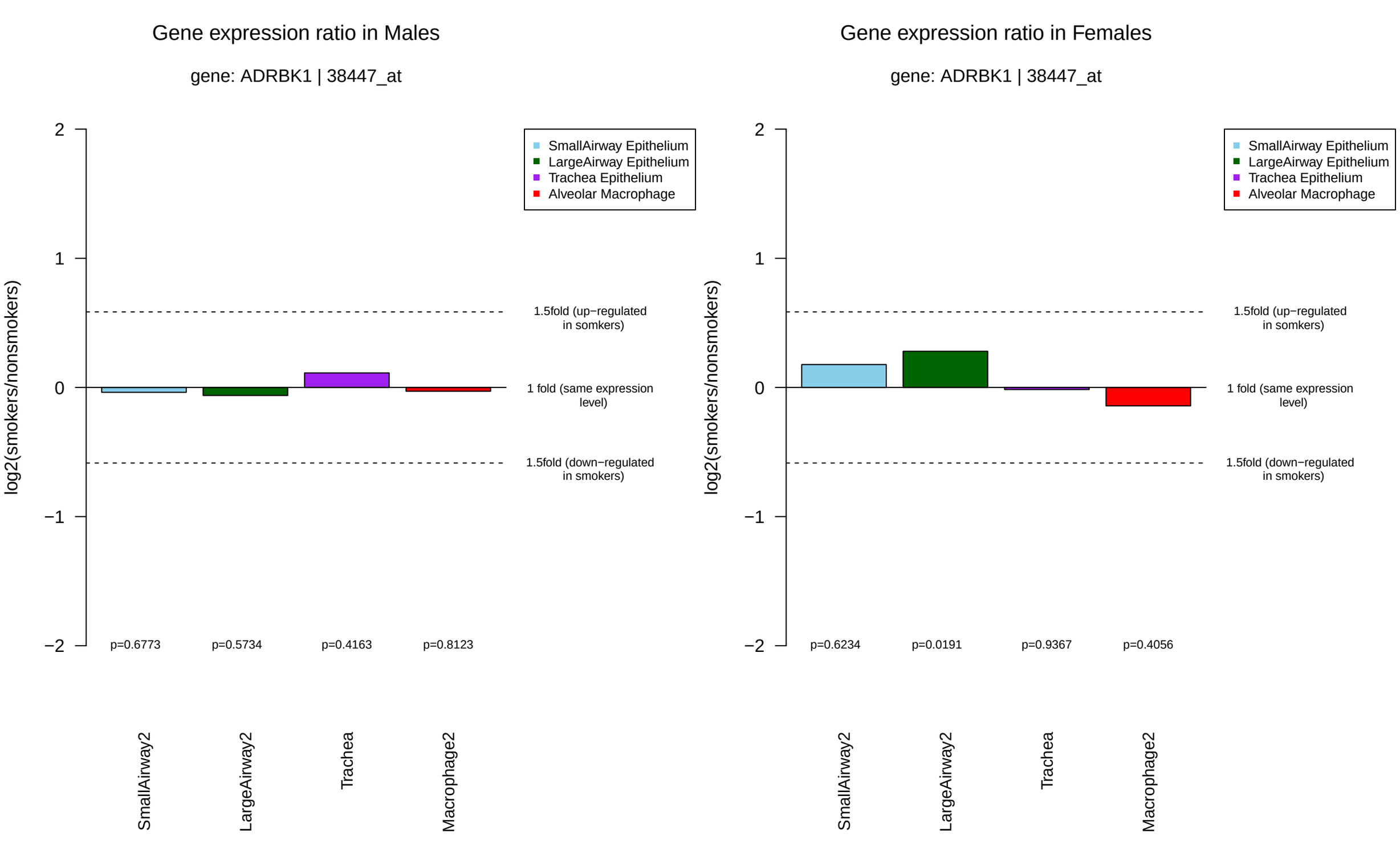
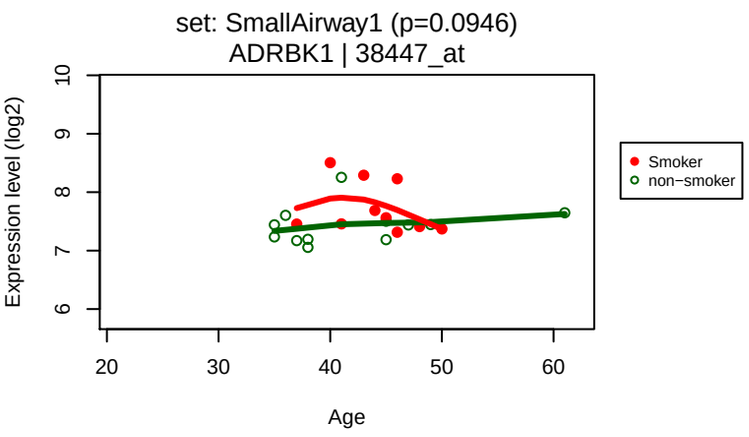
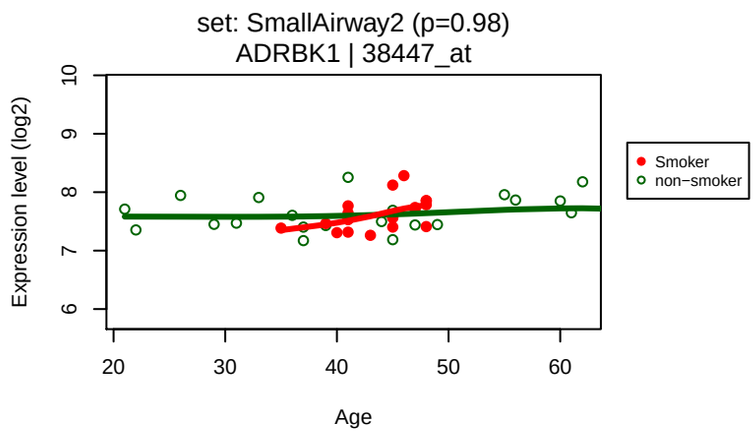
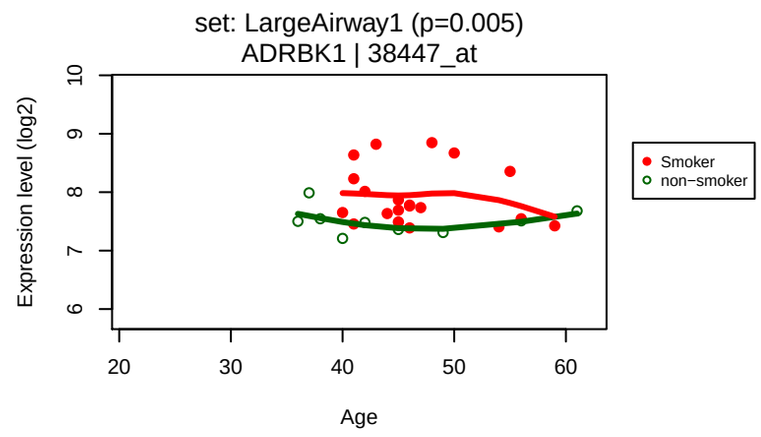

Gene expression of smokers and nonsmokers
gene: ADRBK1 | 38447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0946
p=0.98
p=0.4224
p=0.005
p=0.4232
p=0.7408
p=0.4867
p=0.4779
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

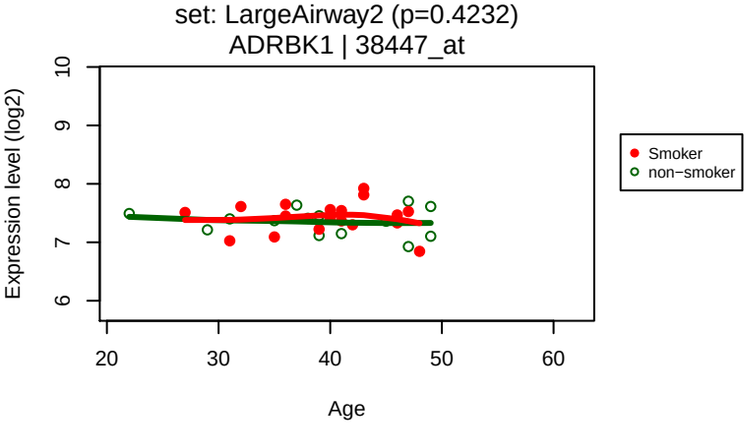

Gene expression ratio (smokers/nonsmokers)
gene: ADRBK1 | 38447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0946
p=0.98
p=0.4224
p=0.005
p=0.4232
p=0.7408
p=0.4867
p=0.4779
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

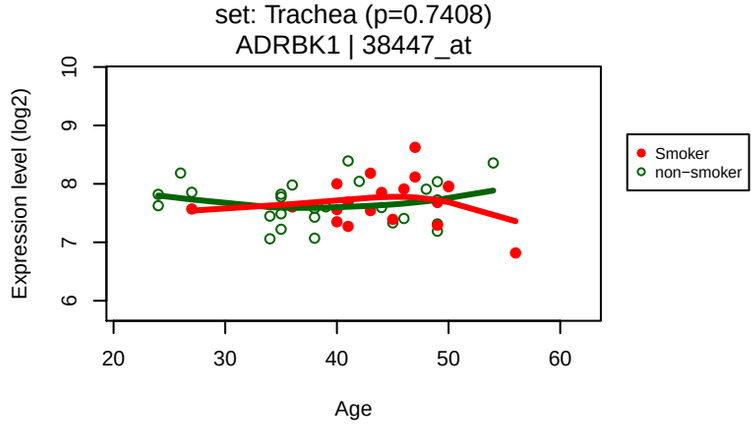

Gene expression ratio in Males
gene: ADRBK1 | 38447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6773
p=0.5734
p=0.4163
p=0.8123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRBK1 | 38447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6234
p=0.0191
p=0.9367
p=0.4056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

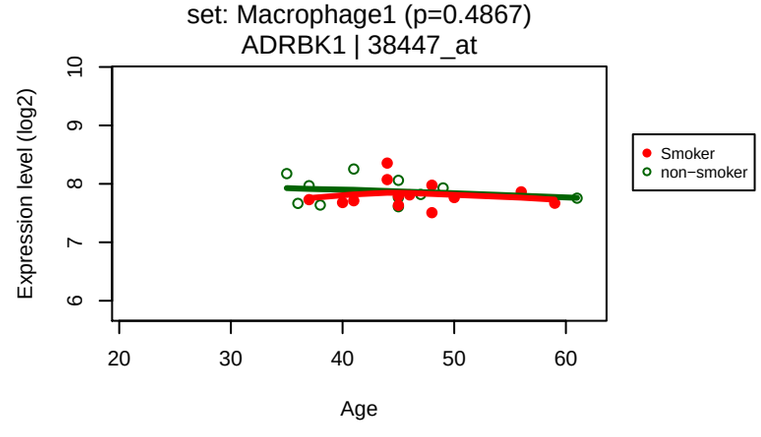

set: SmallAirway1 (p=0.0946)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker

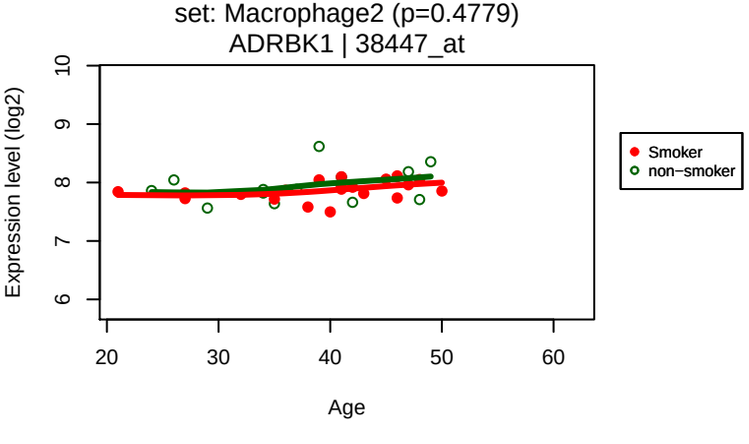

set: SmallAirway2 (p=0.98)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker

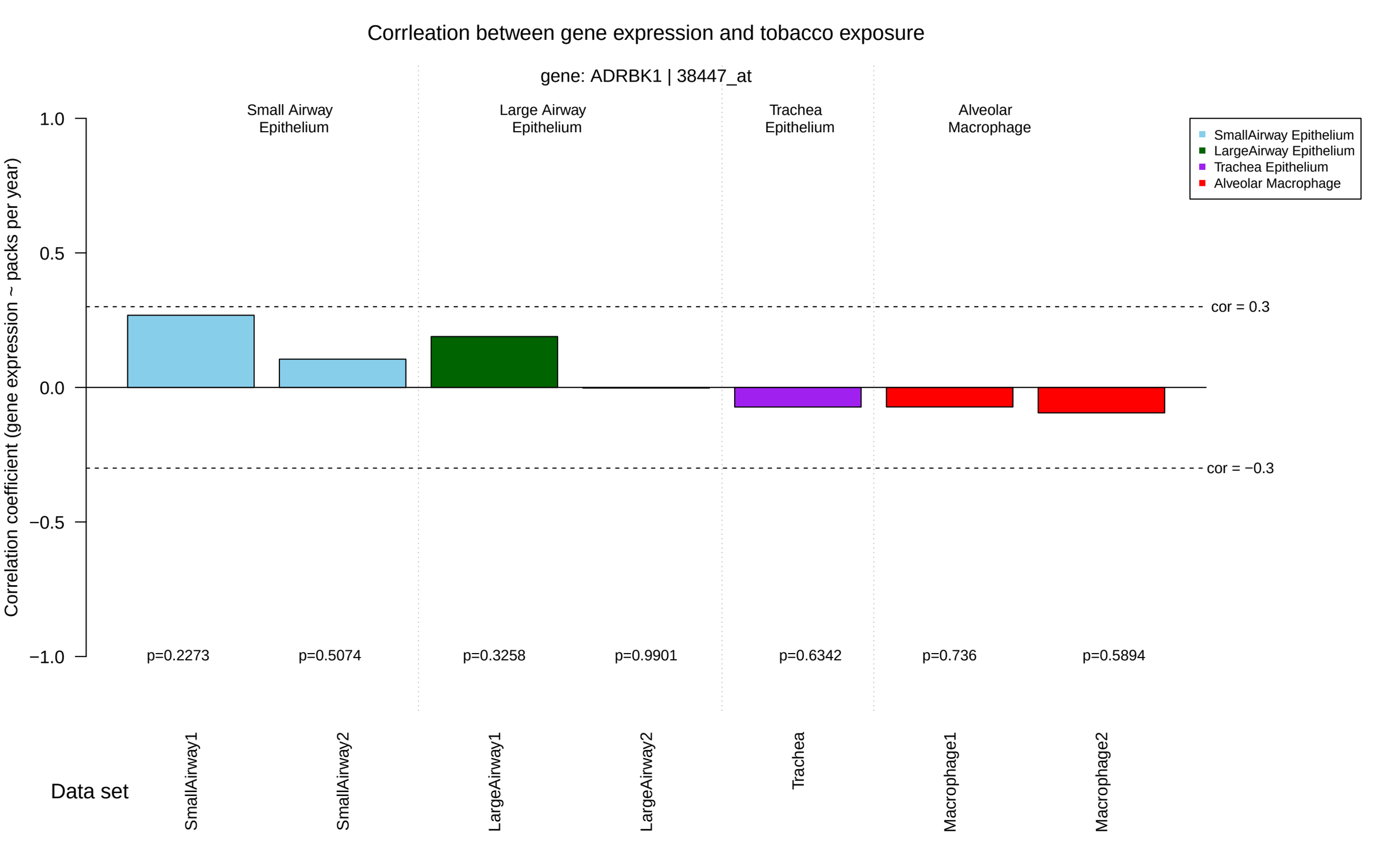

set: LargeAirway1 (p=0.005)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4232)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7408)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4867)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4779)
ADRBK1 | 38447_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRBK1 | 38447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2273
p=0.5074
p=0.3258
p=0.9901
p=0.6342
p=0.736
p=0.5894
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2